Capping the Capacity Market: A Supply Function Equilibrium Analysis of Price Controls in PJM

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Abstract

In January 2025, Pennsylvania Governor Josh Shapiro negotiated a settlement reshaping PJM's capacity market demand curve: a price cap at \$325 per megawatt-day (MW-day) of unforced capacity (UCAP) and a price floor at \$175/MW-day UCAP for the 2026/27 and 2027/28 delivery years, projected to save consumers over \$21 billion. I evaluate this intervention using a Supply Function Equilibrium (SFE) model calibrated to all seven completed Base Residual Auctions (BRAs). Under the maintained assumption of $K \leq 3$ collectively pivotal suppliers, the model's equilibrium lies in an at-cap regime in which the Holmberg boundary condition mechanically fixes the clearing price at the Variable Resource Requirement (VRR) Point (a) value, so the 2026/27 and 2027/28 outcomes are consistent with the model's structural regime rather than an independent forecast of the dollar price. The substantive quantitative result is the binding gap between the VRR Point (a) value (\$329/MW-day in 2026/27) and the settlement cap (\$325/MW-day)—\$4–\$8/MW-day at the Regional Transmission Organization (RTO) level, with larger gaps in constrained Locational Deliverability Areas (LDAs)—and the transition from at-cap to interior clearing between $K = 3$ and $K = 4$. Using the SFE rather than the unconstrained VRR Point (a) as the counterfactual, the implied two-year RTO-level revenue transfer is approximately \$612 million, substantially smaller than the \$21 billion figure. Because both 2026/27 and 2027/28 cleared below their reliability requirements, however, the at-cap outcome is consistent with both the strategic mechanism and physical scarcity, and the counterfactual magnitude is sensitive to which Point (a) represents the pre-settlement benchmark.

1 Introduction

On December 30, 2024, the Commonwealth of Pennsylvania filed a complaint with the Federal Energy Regulatory Commission (FERC) alleging that PJM's capacity auction demand curve was miscalibrated, with its price cap set too high given near term expectations for new entry. The complaint cited PJM's July 2024 Base Residual Auction (BRA) for the 2025/26 delivery year, which cleared at \$269.92/MW-day, an increase of more than 800 percent from the prior year, as evidence that the existing market design imposed unnecessary costs on 65 million consumers across 13 states and the District of Columbia (Office of Governor Josh Shapiro, 2025). Six weeks later, on January 28, 2025, Governor Josh Shapiro announced an agreement with PJM in which the demand curves for the 2026/27 and 2027/28 auctions would be modified to impose a price cap of approximately \$325/MW-day unforced capacity (UCAP) and a price floor of approximately \$175/MW-day UCAP, applied to both the Regional Transmission Organization (RTO) and all constrained Locational Deliverability Areas (LDA) (Office of Governor Josh Shapiro, 2025; Bresler and Keech, 2025). The agreement was projected to save consumers over \$21 billion over the two delivery years.

The settlement reshaped PJM's Variable Resource Requirement (VRR) demand curve by imposing a cap at $\bar{p}_S \approx$ \$325/MW-day below the existing design-level cap at Point (a), and adding a floor at $\bar{p}_f \approx$ \$175/MW-day (Section 3). This paper asks what price PJM's 2026/27 and 2027/28 auctions would have cleared at absent the settlement modification, and what the consequences of suppressing prices to $\bar{p}_S$ are.

I answer the question using a calibrated Supply Function Equilibrium (SFE) model of the PJM BRA. In the model, K symmetric strategic sellers simultaneously submit supply functions against PJM's VRR demand curve, which is publicly known before bidding. Following Klemperer and Meyer (1989) and Green and Newbery (1992), equilibrium is characterized by a first order ordinary differential equation (ODE) system in the sellers' supply functions, integrated backward from the capacity constrained boundary. Equilibrium selection imposes the Holmberg boundary condition, requiring each seller's supply function to equal installed

capacity at the VRR Point (a) cap. This criterion is motivated by Holmberg (2008)'s uniqueness result for stochastic markets, and in the deterministic BRA it is imposed as a selection device. Because the boundary condition forces $p^* = \bar{p}$ whenever it is active, the 2026/27 and 2027/28 at-cap outcomes are consistent with the maintained assumption of $K \leq 3$ strategic sellers but do not independently test the dollar value of the clearing price. The model's substantive content is the regime (at-cap versus interior) and its dependence on K, together with the markup level when the equilibrium is interior. I calibrate the model to seven completed BRA delivery years (2021/22 through 2027/28) using published PJM planning parameters, BRA clearing results, and Independent Market Monitor (IMM) State of the Market cost data. The SFE equilibrium price p^* is the counterfactual clearing price under strategic bidding against the unmodified VRR curve. When the boundary condition is active, p^* equals the design-level Point (a) cap by construction, which is what distinguishes the SFE counterfactual from the VRR Point (a) benchmark used in the settlement's $21 billion savings figure.

2 Literature Review

The supply function equilibrium concept, formalized for the demand-uncertainty case by Klemperer and Meyer (1989), considers an oligopoly in which each firm commits to a schedule mapping prices to quantities before demand is realized. Unlike Bertrand or Cournot competition, the supply function strategy allows each firm to adapt its output to whatever market price emerges after the demand shock is realized. Klemperer and Meyer show that symmetric Nash equilibria are characterized by a first-order differential equation whose solutions form a continuum bounded above by the Cournot schedule and below by the Bertrand schedule (which coincides with the competitive schedule $p = c$ under constant marginal cost, as in the present calibration), with demand uncertainty disciplining firms to choose upward-sloping supply functions.

Green and Newbery (1992) apply the Klemperer-Meyer framework to the England and Wales electricity market, establishing the methodological template I follow: derive the symmetric ODE, integrate backward from the capacity-constrained boundary, and calibrate to observed market structure. Holmberg (2008) shows that in capacity constrained electricity markets, the symmetric SFE is unique when rivals' capacity constraints simultaneously bind with positive probability, tying the ODE boundary condition and selecting a unique equilibrium trajectory. This paper applies that boundary condition as an equilibrium selection device in the deterministic BRA. For the asymmetric extension, Anderson and Hu (2008) develop a discretization-based approximation scheme with provable convergence as the demand-shock grid is refined. Holmberg (2007) establishes existence and uniqueness results for the constant marginal cost, asymmetric capacity case. Rudkevich et al. (1998) provide an early application of supply function analysis to competitive poolco electricity markets, bridging the canonical Klemperer-Meyer theory and applied generation market modeling. The symmetric case applies here and I solve the ODE backward from the Holmberg boundary using the deSolve package in R.

The capacity market design literature originates from the "missing money" problem, where competitive energy markets, subject to regulatory price caps, suppress the scarcity rents that would otherwise finance investment in peaking generation (Joskow and Tirole, 2007; Joskow, 2008). PJM's Variable Resource Requirement curve is a downward-sloping administrative demand schedule that adds elasticity relative to the vertical demand curve it replaced and integrates scarcity pricing into the capacity clearing process (Bowring, 2013). This paper takes the VRR design as given and analyzes how its slope governs strategic markup opportunities under concentrated supply.

Weitzman (1974) shows that the choice between price and quantity regulatory instruments depends on the relative slopes of marginal benefit and marginal cost. When supply is inelastic in the short run, price instruments dominate. Glaeser and Luttmer (2003) demonstrate in the rent control context that binding price controls generate consumption misallocation

qualitatively different from the standard deadweight triangle. The analysis in Section 8 draws on these results to characterize the allocative consequences of the settlement's price band. Cramton and Stoft (2005) emphasize that a central function of the capacity market is to provide a long-run investment signal. A binding cap disrupts this signal precisely when it is most informative.

3 Institutional Background

PJM Interconnection operates the largest competitive wholesale electricity market in North America, coordinating approximately 201,000 megawatts of installed generating capacity across thirteen states and the District of Columbia. Its capacity market, the Reliability Pricing Model (RPM), was introduced in 2007 to procure forward commitments to be available to produce energy during a future delivery year (Bowring, 2013; PJM Interconnection, L.L.C., 2025; Pfeifenberger et al., 2011). A capacity commitment is distinct from the energy itself, which is sold separately in PJM's day-ahead and real-time markets: the capacity payment compensates a resource for standing ready, while energy payments compensate it for actual generation when called upon.

The market exists to address the "missing-money" problem. Regulatory price caps in the wholesale energy market suppress the scarcity rents that would otherwise finance investment in peaking generation, leaving energy revenues insufficient to recover the fixed costs of new entry (Joskow and Tirole, 2007; Joskow, 2008; Cramton and Stoft, 2005). RPM fills the gap with a separate forward instrument that pays for availability and so completes the revenue stream needed for resource adequacy.

The primary procurement mechanism is the Base Residual Auction (BRA), held approximately three years before the start of the delivery year that the auction procures for. Sellers submit offers, PJM clears the auction at a uniform price, and cleared resources receive that clearing price, in dollars per megawatt-day, for every day of the twelve-month delivery year,

paid out as monthly capacity revenue. The 2026/27 delivery year, for example, was procured in the BRA held in July 2025; the clearing price set in that auction governs payments to cleared resources from June 2026 through May 2027. Recent BRAs have run on compressed lead times relative to the canonical three-year forward design (Table 1); the implications of this compression for entry response and for the Shapiro complaint are discussed below.

Capacity is denominated in unforced capacity (UCAP) terms rather than nameplate installed capacity (ICAP). Each resource's accreditation ratio is the fraction of nameplate capacity it is expected to deliver during system stress, accounting for forced outages and, more recently, Effective Load Carrying Capability (ELCC) adjustments for intermittent and energy-limited resources (PJM Interconnection, L.L.C., 2025, 2024). The reference dual-fuel combustion turbine used in PJM's planning parameters has an accreditation of approximately 79%, so a 100 MW nameplate unit clears as 79 MW of UCAP and is paid the clearing price on those 79 MW. Throughout the paper, prices and quantities are stated in UCAP terms unless noted otherwise.

The bidders in a BRA are owners of capacity resources: existing generators (the dominant category by both count and cleared MW), demand response providers, planned new generation, and behind-the-meter distributed resources. Multi-unit ownership is the norm, with large generation owners typically participating with portfolios of dozens to hundreds of resources spread across multiple Locational Deliverability Areas (LDAs). Offers are submitted on a per-resource basis rather than at the firm level, so a single firm enters the auction as a collection of resource-level offers rather than as a single bidder with a portfolio supply schedule. There is no buy-side bidding in the classical sense; PJM acts as the administrative buyer on behalf of Load Serving Entities (LSEs), submitting demand for the RTO and for each LDA through the Variable Resource Requirement (VRR) curve described below.

A resource-level offer is a sealed price-quantity pair, denominated in dollars per megawatt-day of UCAP, that specifies the minimum payment at which the resource is willing to commit. Conceptually the offer is a going-forward cost: the per-MW payment the resource needs

above expected energy and ancillary-service revenues to remain available during the delivery year. For an existing resource this is a short-run avoidable cost (operations, maintenance, and forgone alternative use); for new entry it is the annualized fixed cost net of expected energy revenues. The PJM Independent Market Monitor (IMM) computes a resource-specific Net Avoidable Cost Rate (Net ACR) that serves both as a default offer cap under mitigation and as the cost benchmark referenced in the calibration of Section 5 (Monitoring Analytics, LLC, 2026).

A cleared resource takes on a Capacity Performance obligation: it must be available to produce energy when PJM declares Performance Assessment Hours during the delivery year, and non-performance triggers a charge calibrated to a multiple of Net CONE (PJM Interconnection, L.L.C., 2025). The performance obligation completes the meaning of the capacity commitment and ties the forward payment to delivery-year availability. The penalty mechanics are not modeled in this paper; the SFE setup in Section 4 treats committed capacity as delivered.

Clearing is a one-sided sealed-bid uniform-price auction. PJM aggregates the resource-level offers in each LDA into an upward-stepping supply schedule and intersects it with the LDA's VRR demand curve; the intersection price is the LDA clearing price, and every resource whose offer lies at or below that price clears its full offered quantity at the uniform clearing price. Clearing is solved jointly across the RTO and the constrained sub-LDAs subject to Capacity Emergency Transfer Limit (CETL) constraints that bound the imports each LDA can receive from the rest of the system (PJM Interconnection, L.L.C., 2024). When a CETL constraint binds, the LDA's clearing price separates from the RTO price and the constrained LDA pays a locational premium.

In auction-theoretic terms, the BRA is a one-sided, sealed-bid, uniform-price, multi-unit, multi-product mechanism with a publicly known administrative demand schedule. The multi-unit dimension reflects each seller's portfolio of resources within an LDA; the multi-product dimension reflects the joint clearing across LDAs subject to CETL. The publicly known

demand curve, the VRR, is the primitive on which the supply function equilibrium analysis in Sections 4 and 6 rests, and it is the object that the 2025 Shapiro settlement modified.

The empirical panel covers the RTO and sixteen sub-LDAs across delivery years 2021/22 through 2027/28.

The VRR curve is the administrative demand schedule presented to sellers in each BRA (PJM Interconnection, L.L.C., 2025). Under the old design (used for delivery years 2021/22 through 2025/26), it is defined by three anchor points: Point (a), the price cap at $1.5 \times$ Net CONE, Point (b) at $0.75 \times$ Net CONE at the reliability requirement, and Point (c) at zero price above the reliability requirement (Bowring, 2013; Pfeifenberger et al., 2011). The Net Cost of New Entry (Net CONE) is the annualized cost of a reference generating unit, net of projected energy and ancillary service revenues, computed independently for each LDA (PJM Interconnection, L.L.C., 2025, 2024). FERC approved a restructured VRR curve design in 2025 (Federal Energy Regulatory Commission, 2025), taking effect with the 2026/27 delivery year. The redesign added a fourth anchor point, creating an explicit price floor at approximately \$177–\$179 per MW-day and eliminating the collapse to zero price for surplus capacity.

Table 1 documents the BRA timeline and RTO clearing outcomes for all seven completed auctions in the sample. Both auctions conducted under the new VRR design (2026/27 and 2027/28) cleared at exactly the Point (a) price cap, motivating the SFE analysis in Sections 4–6.

Table 1: PJM Base Residual Auction Timeline and RTO Clearing Outcomes, 2021/22 to 2027/28

Delivery Year	BRA Date	Lead (mo.)	Clearing ($/MW-day)	Net CONE ($/MW-day)	VRR Design	At Cap
2021/22	May 2018	37.0	$140.00	$321.57	Old	No
2022/23	May 2021	12.5	$50.00	$260.50	Old	No
2023/24	July 2022	11.0	$34.13	$274.96	Old	No
2024/25	December 2022	17.8	$28.92	$293.19	Old	No
2025/26	July 2024	10.1	$269.92	$228.81	Old	No
2026/27	July 2025	10.2	$329.17	$212.14	New	Yes
2027/28	December 2025	17.7	$333.44	$242.52	New	Yes

Note: Prices in nominal dollars per MW-day. "At Cap" indicates the RTO clearing price equaled the Point (a) price cap. Source: PJM BRA results and planning parameters.

PJM's primary market power mitigation mechanism is the Three Pivotal Supplier (TPS) test. Supplier i is pivotal in LDA ℓ if removing i's capacity would cause aggregate supply to fall short of the reliability requirement. The Residual Supply Index for k suppliers is defined as

$$\mathrm{RSI}_k^\ell = \frac{\text{Total capacity in LDA } \ell \; - \; \text{Capacity of top-}k \text{ sellers}}{\text{Reliability requirement for LDA } \ell}. \tag{1}$$

Mitigation is triggered when $\mathrm{RSI}_1 \leq 1$ or $\mathrm{RSI}_3 \leq 1$, capping affected suppliers' offers at their Net Avoidable Cost Rate (ACR) (Bowring, 2013; Monitoring Analytics, LLC, 2026).

Table 2 summarizes RTO-level TPS test outcomes for the six delivery years with complete IMM data. RSI_3 is below 1 in every year, ranging from 0.62 to 0.73, so the mitigation threshold is met throughout the sample and concentration is persistent. The IMM also publishes LDA-level RSI values in its annual State of the Market reports; the cross-sectional snapshots corroborate this RTO pattern and document individual pivotality in several small constrained LDAs (Monitoring Analytics, LLC, 2026).

Table 2: TPS Test Results at the RTO Level: Residual Supply Indices by Delivery Year

Delivery Year	RSI_1	RSI_3	Participants	All Pivotal
2022/23	0.81	0.73	130	Yes
2023/24	0.78	0.68	134	Yes
2024/25	0.77	0.64	133	Yes
2025/26	0.82	0.62	128	Yes
2026/27	0.82	0.64	153	Yes
2027/28	0.82	0.63	155	Yes

Note: "All Pivotal" indicates $RSI_1 < 1$ for each of the three largest sellers individually. Source: PJM IMM State of the Market reports, various years.

The high BRA clearing prices of 2025/26 and the anticipation of similarly elevated prices in subsequent auctions triggered political intervention that directly modified the demand curve for two delivery years. On December 30, 2024, the Office of Governor Josh Shapiro filed a complaint against PJM's tariff with the Federal Energy Regulatory Commission under Section 206 of the Federal Power Act. The complaint's legal theory was that the VRR Point (a) price cap was set too high given lower expectations for new entry in the near term (Bresler and Keech, 2025). Because compressed auction lead times (Table 1) limited the ability of new generation to respond to high prices within upcoming delivery periods, the complaint argued that high prices would not produce efficient entry and instead represented an unjust and unreasonable rate. The proposed remedy was to lower the cap to $1.5 \times$ Net CONE and remove reliance on Gross CONE in the Point (a) calculation.

Following confidential settlement discussions, PJM and the Governor's office announced an agreement in principle on January 28, 2025 (Office of Governor Josh Shapiro, 2025). The settlement was presented to the PJM Markets Committee on February 7, 2025, and PJM indicated a target filing date of February 14, 2025, with the 2026/27 BRA scheduled for July 2025 and the 2027/28 BRA for December 2025 (Bresler and Keech, 2025). The

agreement imposes a price band on the demand curves for both delivery years: a price cap at approximately \$325/MW-day UCAP (Unforced Capacity), equivalent to \$256.75/MW-day in Installed Capacity (ICAP) terms at the assumed 79% accreditation of the dual-fuel combustion turbine reference unit, and a price floor at approximately \$175/MW-day UCAP (\$138.25/MW-day ICAP), below which all available capacity clears regardless of the VRR slope. The settlement cap is denoted as $\bar{p}_S \approx$ \$325/MW-day (UCAP) and the floor as $\bar{p}_f \approx$ \$175/MW-day (UCAP) to distinguish them from the existing VRR Point (a) cap $\bar{p}$. Both values are indexed to the accreditation of the reference unit and may differ slightly from these round numbers when final Effective Load Carrying Capability (ELCC) values are determined.

The settlement does not impose an external price constraint on an otherwise unchanged auction. Instead, it modifies the shape of the VRR demand curve itself (Bresler and Keech, 2025). At prices above $\bar{p}_S$, the demand curve is flat (no capacity with offers above the cap can clear). Between $\bar{p}_S$ and $\bar{p}_f$, the existing VRR sloped segment applies. At prices at or below $\bar{p}_f$, the demand curve extends horizontally to clear all available capacity at the floor price. This mechanism applies identically to the RTO demand curve and to every LDA demand curve. Clearing results continue to respect CETL transmission limitations (Bresler and Keech, 2025; Federal Energy Regulatory Commission, 2025). The pre-existing Point (a) cap $\bar{p}$ is calibrated to investment incentives (approximately $1.5 \times$ Net CONE), while the settlement cap $\bar{p}_S$ is lower and is motivated by near-term consumer cost concerns. Table 1 shows that $\bar{p} =$ \$329.17/MW-day in 2026/27, only \$4.17 above $\bar{p}_S =$ \$325/MW-day; the gap is larger in constrained LDAs where local Point (a) caps substantially exceeded \$325 in prior years.

4 Model

The BRA is modeled as a static game in which sellers simultaneously submit supply functions before the auction clears. This formulation follows Klemperer and Meyer (1989), whose

original framework admits demand uncertainty, with Green and Newbery (1992) applying it to electricity markets under deterministic demand. The VRR curve is deterministic, so I use the Klemperer-Meyer ODE as an equilibrium selection device rather than as the equilibrium concept itself. Timing: (1) PJM publishes the VRR parameters for the upcoming delivery year; (2) sellers submit supply functions; (3) PJM clears the auction at the uniform price where aggregate supply meets the VRR curve; (4) sellers deliver committed capacity during the delivery year.

Assumption 1. *There are $K \geq 2$ strategic sellers indexed $i = 1, \ldots, K$ and a competitive fringe. Each seller i holds a fixed capacity portfolio $\bar{q}_i > 0$ measured in megawatts. A supply function $S_i : [0, \bar{p}] \to [0, \bar{q}_i]$ is a nondecreasing, twice continuously differentiable mapping from price to quantity. Each seller simultaneously commits to a supply function before the auction clears.*

Assumption 2. *The competitive fringe bids at its avoidable cost rate $c_f > 0$. Fringe supply is $S_f(p) = 0$ for $p < c_f$ and $S_f(p) = \bar{Q}_f$ for $p \geq c_f$, where $\bar{Q}_f > 0$ is total fringe capacity. The fringe is price taking and its behavior is common knowledge.*

Assumption 3. *Each strategic seller has a constant marginal cost $c > 0$ for output $q \leq \bar{q}_i$ and infinite marginal cost above capacity. For the symmetric calibration I set c equal to the sector average avoidable cost rate published by the PJM IMM.*

The market clears at the price p^* satisfying

$$\sum_{i=1}^{K} S_i(p^*) + S_f(p^*) = D(p^*), \tag{2}$$

where $D(\cdot)$ is the VRR demand schedule described below. Every seller receives the same clearing price p^* for its committed capacity. The residual demand facing seller i at price p is

$$R_i(p) \;=\; D(p) - \sum_{j \neq i} S_j(p) - S_f(p). \tag{3}$$

In equilibrium, $S_i(p^*) = R_i(p^*)$. Because $D(\cdot)$ and all rivals' strategies are publicly known before the auction, each seller's residual demand is deterministic. The boundary condition below selects among equilibria.

PJM's publicly announced VRR curve is a piecewise linear, downward sloping demand schedule. Under the old design (delivery years 2021/22 to 2025/26) it is parameterized by three anchor points (p_a, q_a), (p_b, q_b), and $(0, q_c)$:

$$D(p) = \begin{cases} q_a & p \geq p_a \\ q_a + \dfrac{q_b - q_a}{p_b - p_a}(p - p_a) & p_b \leq p < p_a \\ q_b + \dfrac{q_c - q_b}{0 - p_b}(p - p_b) & 0 \leq p < p_b, \end{cases} \tag{4}$$

where $p_a = 1.5 \times$ Net CONE is the price cap and $p_b = 0.75 \times$ Net CONE is the reliability requirement price. The quantities $q_a < q_b < q_c$ are the corresponding capacity levels, with q_b equal to the reliability requirement. Because $D(p) = q_a$ for all $p \geq p_a$, the VRR curve is effectively inelastic above the cap.

Under the new design (delivery years 2026/27 onward), FERC Order ER25-1357 introduces a fourth anchor point (p_f, q_f) and (p_f, q_d) with $q_f > q_b$, creating a flat floor segment at price $p_f \approx$ \$177–\$179/MW-day. The new schedule is:

$$D(p) = \begin{cases} q_a & p \geq p_a \\ q_a + \dfrac{q_b - q_a}{p_f - p_a}(p - p_a) & p_f \leq p < p_a \\ q_d & 0 \leq p < p_f. \end{cases} \tag{5}$$

The flat floor segment at p_f means that clearing prices cannot fall below p_f along the VRR, changing the elasticity profile that sellers face and thus the equilibrium markup. The comparative static in Section 6 isolates this design change. The slope $D'(p)$ is piecewise constant and satisfies

Assumption 4. *$D'(p) < 0$ on the sloped segments; $D'(p) = 0$ on the flat segments (at and above p_a, and below p_f under the new design). The function $D(\cdot)$ is known to all players before the auction.*

Seller i chooses $S_i(\cdot)$ to maximize profit. Taking rivals' supply functions as fixed, the profit at any market clearing price p is $\pi_i = p \cdot S_i(p) - c \cdot S_i(p) = (p - c) \cdot S_i(p)$, since marginal cost is constant at c. The seller's problem is equivalent to maximizing profit along its residual demand curve $R_i(\cdot)$. At price p, if seller i considers increasing its offered quantity by $\mathrm{d}q$, the market clearing price must fall by $\mathrm{d}p$ to preserve equation (2):

$$\mathrm{d}p \;=\; \frac{\mathrm{d}q}{D'(p) - \sum_{j \neq i} S_j'(p) - S_f'(p)}. \tag{6}$$

Since $D'(p) < 0$ and $S_j'(p) \geq 0$, the denominator is strictly negative, so any increase in supply lowers the clearing price. The seller's marginal revenue from increasing quantity at price p is:

$$\mathrm{MR}_i(p) \;=\; p + \frac{S_i(p)}{D'(p) - \sum_{j \neq i} S_j'(p) - S_f'(p)}. \tag{7}$$

Setting marginal revenue equal to marginal cost c yields the *supply function equilibrium condition*:

Proposition 1. *In a supply function equilibrium, each seller i's strategy $S_i(\cdot)$ satisfies*

$$S_i(p) \;=\; -\Big(p - c\Big)\Big[D'(p) - \textstyle\sum_{j \neq i} S_j'(p) - S_f'(p)\Big] \tag{8}$$

at every price p at which the market can clear. This is a system of K coupled first-order ordinary differential equations in the supply functions $\{S_1, \ldots, S_K\}$.

Equation (8) is the K-seller generalization of the first-order condition derived by Green and Newbery (1992) for the symmetric duopoly (their equation 3) and of the general condition in Anderson and Hu (2008) (their equation 1). Its economic content is straightforward: seller

i equates its residual-demand markup $(p - c)$ to the inverse of the residual demand elasticity at that price. When more sellers compete (K large) or the VRR slope is steep (large $|D'|$), the system requires a smaller markup at each price, pushing the equilibrium supply function toward the competitive schedule.

The ODE system (8) generally admits a continuum of solutions. Klemperer and Meyer (1989) show that under demand uncertainty with bounded support, the symmetric SFE trajectories form a connected interval bounded above by the Cournot schedule and below by the Bertrand schedule (which coincides with $p = c$ under constant marginal cost). Any trajectory between these stationary solutions is a Nash equilibrium, so prediction and comparative statics require a selection criterion. I use the following boundary condition, which applies in the capacity auction setting:

Proposition 2 (Holmberg 2008)**.** *Suppose (i) sellers are symmetric, (ii) demand is perfectly inelastic below the price cap $\bar{p}$ and zero above, (iii) seller cost is strictly convex, twice continuously differentiable, with marginal cost below the cap at full capacity, and (iv) the probability that aggregate demand reaches the capacity ceiling is strictly positive. Then there exists a unique symmetric supply function equilibrium. In this equilibrium each seller's supply function satisfies the boundary condition*

$$S_i(\bar{p}) \;=\; \bar{q}_i \quad \textit{for all } i. \tag{9}$$

Holmberg's original proof is under demand uncertainty. In the deterministic BRA studied here I impose (9) as an equilibrium selection device rather than deriving it from the theorem.

The intuition: when rivals' capacity constraints simultaneously bind, seller i faces perfectly inelastic residual demand and its optimum is to price at the cap $\bar{p}$. Any lower price sacrifices revenue without gaining quantity, and any higher price is infeasible. This forces the supply function through $(\bar{p}, \bar{q}_i)$, pinning the arbitrary constant in the ODE. In the PJM BRA, $\bar{p}$ equals the VRR Point (a) price, and Assumption 4 gives $D'(\bar{p}) = 0$ (the VRR is flat at the

cap). The at-cap clearing outcomes in 2026/27 and 2027/28 (Table 1) confirm the boundary condition is active in those years, consistent with the Holmberg (2008) selection criterion. PJM's reliability target requires procuring nearly all installed capacity, so binding-capacity outcomes are the empirically relevant case. I therefore integrate (8) *backward* from the boundary condition $S_i(\bar{p}) = \bar{q}_i$.

For calibration and comparative statics, I begin with the symmetric case in which all K strategic sellers are identical with $\bar{q}_i = \bar{q}$ and $c_i = c$ for all i. In a symmetric Nash equilibrium, $S_i(p) = s(p)$ for all i. Substituting into equation (8):

$$s(p) \;=\; -\Big(p - c\Big)\Big[D'(p) - (K-1)\, s'(p) - S_f'(p)\Big]. \tag{10}$$

Under Assumption 2, $S_f'(p) = 0$ almost everywhere. Rearranging (10) for $s'(p)$ yields the symmetric SFE ordinary differential equation

$$s'(p) \;=\; \frac{1}{K-1}\left[D'(p) + \frac{s(p)}{p-c}\right], \quad p \in [c,\, \bar{p}), \tag{11}$$

with boundary condition $s(\bar{p}) = \bar{q}$. This is a first-order, nonautonomous ODE in the single unknown function $s(\cdot)$. It is the K-firm generalization of Green and Newbery's equation (3), to which it reduces when $K = 2$ and $S_f \equiv 0$.

Equation (11) has a direct economic interpretation. The $D'(p)$ term captures the direct effect of price on demand: when demand is more responsive (larger $|D'|$), the equilibrium supply function is steeper in price. The $s(p)/(p-c)$ term is a markup-boundary feedback: as $p \to c$ from above, it diverges and keeps $s'(p)$ bounded, enforcing the lower boundary and ruling out below-cost offers. Well above c, the feedback partially offsets $D'(p)$, so the supply function is steeper than a linear interpolation between c and $\bar{p}$ would predict. More competitors (larger K) divide the denominator and flatten each firm's supply function toward the competitive benchmark.

The equilibrium clearing price p^* is found by solving

$$K \cdot s(p^*) + S_f(p^*) = D(p^*) \tag{12}$$

for p^* after integrating (11) backward from $\bar{p}$. The implied price-cost margin (Lerner index) is

$$L \;=\; \frac{p^* - c}{p^*} \tag{13}$$

which measures the equilibrium price-cost margin reported in Section 6. The competitive benchmark corresponds to $c = p^*$ ($L = 0$), attained as K approaches infinity or as $|D'(p)|$ approaches infinity (perfectly elastic demand). The Cournot upper bound on the markup is $L^{\text{Cournot}} = 1/(K \cdot |D'(p^*) \cdot p^*/D(p^*)|)$, the standard inverse elasticity rule evaluated at market demand (which equals the residual demand elasticity in the symmetric case).

Treating the three largest PJM sellers as symmetric is a tractability assumption. It reduces the K-dimensional coupled ODE system (8) to the scalar ODE (11), which is calibratable from publicly available aggregate data. This is the standard approach in the calibrated SFE literature (Green and Newbery, 1992; Rudkevich et al., 1998). The symmetric Lerner index approximates the capacity-weighted average markup; asymmetric extensions are left for future work. Equation (11) is solved numerically with a fourth-order Runge-Kutta integrator using fixed step size $h = 0.05\,\$/\text{MW-day}$, integrating backward from $s(\bar{p}) = \bar{q}$.

5 Calibration

The calibration draws on four publicly available datasets produced by PJM and the IMM. PJM publishes VRR curve anchor points, Net CONE, CETL, and reliability requirements for every LDA in advance of each BRA. I compiled planning parameter files for delivery years 2021/22 through 2028/29, covering both the three-point (old) and four-point (new) curve designs. PJM releases clearing prices and quantities by LDA after each auction, and my

sample covers seven completed auctions (2021/22 through 2027/28) verified against PJM press releases. Residual Supply Index values and participant counts are drawn from the IMM's annual State of the Market reports (Volumes 2022 and 2025). RSI data is available for a subset of LDAs and delivery years, as documented in Table 2. Lead time is defined as the number of calendar months between the BRA settlement date and the first day of the delivery year (Table 1). The four datasets are joined at the LDA delivery year level to produce a master panel with 110 observations spanning 17 LDAs and seven delivery years (2021/22 through 2027/28).

Table 3 reports the RTO VRR parameters for the six delivery years with complete BRA data. Under the old design, Points (a) and (b) equal 1.5 × Net CONE and 0.75 × Net CONE by construction, so the ratio $p_a/p_b = 2$ holds exactly in every old-design year except 2025/26. In that year, a transitional administrative cap fixed Point (a) at \$451.61, above the 1.5 × Net CONE multiple of \$343.22; VRR anchor points are fixed at planning-parameter publication and may diverge from the final Net CONE. I use the published anchor points throughout. Under the new design, Points (b) and (c) collapse to the floor price of \$177–\$179/MW-day, and Point (a) is set by FERC Order ER25-1357. The Point (a) values for 2026/27 (\$329.17) and 2027/28 (\$333.44) are not simple Net CONE multiples; the settlement cap of \$325/MW-day therefore binds \$4.17 and \$8.44 below them.

Table 3: RTO VRR Curve Parameters, 2022/23–2027/28

Delivery Year	Net CONE ($/MW-day)	Pt. (a) Price	Pt. (a) MW	Pt. (b) Price	Pt. (b) MW	VRR Design
2022/23	$260.50	$390.75	136,076	$195.38	139,656	Old
2023/24	$274.96	$412.44	135,914	$206.22	139,473	Old
2024/25	$293.19	$439.79	138,341	$219.89	141,910	Old
2025/26	$228.81	$451.61	133,554	$171.61	137,160	Old
2026/27	$212.14	$329.17	134,887	$177.24	136,361	New
2027/28	$242.52	$333.44	141,681	$179.55	143,218	New

Note: Prices in nominal dollars per MW-day; quantities in MW. The 2025/26 Point (a) price exceeds $1.5 \times$ Net CONE due to a transitional administrative cap. Source: PJM Planning Parameters.

The PJM IMM reports LDA-level Herfindahl-Hirschman Index (HHI) values in its annual State of the Market reports, and concentration is consistently high. The RTO HHI exceeds 2,500 in every sample year, and most constrained LDAs exceed 4,000. The Department of Justice classifies markets above 2,500 as highly concentrated (citation needed).

The key calibration input is the number of strategic sellers, K. I calibrate K against the Residual Supply Index with threshold three (RSI_3). $\text{RSI}_3 < 1$ means the three largest sellers are jointly pivotal: no competitive outcome is supportable without their participation. Across the six sample years with IMM RSI data, the RTO RSI_3 is below 1 (0.62–0.73; Table 2), which supports treating up to three large sellers as jointly pivotal. The 2026/27 at-cap outcome is also consistent with $K \leq 3$: the model predicts interior (below-cap) clearing with four or more symmetric sellers in that year (Section 6). Cross-sectional LDA snapshots in the 2022/23 IMM State of the Market show $\text{RSI}_1 < 1$ for each of the top three sellers in every constrained LDA, supporting $K = 3$ over $K = 2$ as the baseline.[1] I therefore set $K = 3$ as the baseline strategic count and examine sensitivity via the K comparative static in Section 6.

[1]The same 2022/23 snapshot reports single-supplier pivotality ($\text{RSI}_1 = 0$) in BGE and DEOK (Monitoring Analytics, LLC, 2026); a symmetric $K = 3$ calibration understates local market power in these cases.

Because RSI data is available only at the RTO level for most delivery years, I impose the same $K = 3$ structure on each LDA calibration and let cross-LDA variation enter through the VRR parameters and the CETL-based fringe capacity. This homogeneity is an acknowledged limitation. Where a single supplier is pivotal, the effective K is closer to one and the symmetric $K = 3$ calibration understates local market power. Conversely, in LDAs with $\text{RSI}_3 > 1$, three suppliers are not jointly pivotal and $K = 3$ overstates concentration. LDA-level Lerner indices should therefore be read as applying a common structural assumption to heterogeneous local conditions, not as LDA-specific estimates.

The ACR for each strategic seller is set to $c = \$150$/MW-day, the combined-cycle sector average reported in the 2025 IMM State of the Market (Monitoring Analytics, LLC, 2026, Table 7-38, \$149.32/MW-day, rounded). Combined-cycle units are the dominant marginal gas technology in PJM's current resource mix, so this sector average serves as a technology proxy for the generic strategic seller's going-forward cost. PJM applies the TPS offer cap itself on a unit-specific basis using each resource's Net ACR (Monitoring Analytics, LLC, 2026, 2025); plant-level cost data for a more granular calibration are not publicly available. The symmetric-firm assumption sets all $K = 3$ sellers to the same c and abstracts from cost and capacity heterogeneity.

The ACR proxies opportunity cost rather than measuring it directly. True avoidable cost depends on projected energy-market revenues, fuel hedges, maintenance schedules, and the firm's financial structure—none observable at the portfolio level. The sector-average ACR therefore abstracts from cost heterogeneity across strategic sellers. Table 6 varies c over \$100–\$200/MW-day at $K = 3$; the qualitative settlement-binding finding is robust across this range.

The range also serves as a bracketing device. A uniform ACR of \$100/MW-day yields an upper Lerner bound of 0.70 in the 2026/27 at-cap equilibrium; a uniform ACR of \$200/MW-day yields a lower bound of 0.39. Heterogeneous-cost configurations with sector-average ACR in this range need not produce a Lerner index strictly between these symmetric bounds—the

symmetric SFE is not a proven envelope. The bracketing analysis nevertheless establishes that the central Lerner estimate of 0.54 at $c = \$150$ is not an artifact of a narrow cost assumption.

Fringe supply is modeled as an inelastic block offered at c. At the RTO level, fringe capacity equals total installed capacity minus the three largest sellers' capacity, scaled by γ to explore fringe-size sensitivity. For individual LDA calibrations, CETL—the maximum deliverable import from the rest of PJM into the LDA—serves as the fringe proxy. LDA-level CETL values by delivery year are drawn from the PJM planning parameter files described in Section 5.

I calibrate and solve the model for all seven completed BRA delivery years (2021/22 through 2027/28), spanning both VRR design regimes and the full range of observed market conditions. Five auctions (2021/22 through 2025/26) were conducted under the old three-point VRR design; two (2026/27 and 2027/28) under the new four-point design with an explicit price floor. The SFE is a static game in which sellers submit supply functions against the VRR curve; the auction mechanism is structurally identical across all seven years regardless of lead time, so no year is excluded on timing grounds.

6 Results

Table 4 presents the baseline symmetric SFE results for all seven completed BRA delivery years, calibrated with $K = 3$ strategic sellers and $c = \$150$/MW-day (the sector-average ACR for combined-cycle generators from the 2025 IMM State of the Market). For each year, the table reports the model-predicted equilibrium price p^*, the observed BRA clearing price, and the implied price-cost margin (Lerner index) $L = (p^* - c)/p^*$.

The central finding is that the Shapiro cap is *binding*: the SFE equilibrium for 2026/27, $p^* = \$329$/MW-day, exceeds the settlement cap $\bar{p}_S = \$325$/MW-day by \$4/MW-day at the RTO level. For constrained LDAs, the binding gap is larger (\$115–\$141/MW-day based on

Table 4: Baseline SFE Equilibrium: $K = 3$, $c = \$150$/MW-day

Year	Design	$\bar{p}$	$\bar{p}_S$	p^*	p_{actual}	Margin	L	Outcome
2021/22	Old	\$482	—	\$405	\$140	+6.8%	0.63	Interior
2022/23	Old	\$391	—	\$310	\$50	+9.2%	0.52	Interior
2023/24	Old	\$412	—	\$330	\$34	+9.9%	0.55	Interior
2024/25	Old	\$440	—	\$358	\$29	+11.7%	0.58	Interior
2025/26	Old	\$452	—	\$426	\$270	+1.6%	0.65	Interior
2026/27	New	\$329	\$325	\$329	\$329	−0.2%	0.54	At cap
2027/28	New	\$333	\$325	\$333	\$333	−4.7%	0.55	At cap

Notes: $\bar{p}$: VRR Point (a) price cap. $\bar{p}_S$: Shapiro settlement cap (\$325/MW-day UCAP); "—" indicates the settlement does not apply. Prices in \$/MW-day. Margin = $(Q_{\text{cleared}} - Q_{\text{reliability req}})/Q_{\text{reliability req}}$; negative values indicate the auction cleared below the reliability requirement. "Interior": equilibrium falls strictly inside $(c, \bar{p})$; "At cap": $S_i(\bar{p}) = \bar{q}_i$ for all i. Source: PJM BRA clearing data and SFE model.

2025/26 clearing prices of \$444–\$466/MW-day; Section 6).

The 2026/27 result is the most informative for assessing the maintained model. The Holmberg boundary condition selects $p^* = \bar{p} = \$329.17$/MW-day for $K = 3$. It is important to be transparent about what this means: the matching of the observed clearing price is a *mechanical* consequence of the boundary condition, not an independent forecast. Once $K = 3$ is assumed and the 2026/27 capacity margin (−0.2%) is tight enough to trigger the at-cap regime, the ODE solution is forced to $\bar{p}$ by construction. The dollar amount \$329.17 is simply the VRR Point (a) price read from the planning parameters. The substantively informative content is the *binary* at-cap versus interior outcome: for $K \geq 4$, the model predicts interior clearing strictly below the cap (see Section 6), so the observed at-cap outcome is consistent with $K \leq 3$ as the maximal effective seller count in 2026/27. The model does not confirm the price. It confirms the structural regime.

For all five old-design years, p^* substantially exceeds the observed clearing price. The gap is largest in well-supplied years where TPS mitigation was broadly applied (e.g., $p^* = \$330$ vs. $p_{\text{actual}} = \$34$ in 2023/24, with $\text{RSI}_3 = 0.68$ and capacity margin 9.9%). The gap narrows in 2025/26 ($p^* = \$426$ vs. $p_{\text{actual}} = \$270$), reflecting a capacity margin of only 1.6% and a price spike that occurred even under partial mitigation. This pattern is consistent across all

old-design years: the SFE predicts the *unmitigated* equilibrium, while actual prices reflect the TPS screen.

This distinction shapes how the 2026/27 and 2027/28 results should be read. The SFE equilibrium p^* describes what the auction would clear at under strategic bidding when no binding mitigation cap is imposed. The observed clearing price equals the minimum of p^* and the TPS-mitigated offer cap (approximately $c \cdot (1+m)$, where m is the IMM's mitigation multiplier); in old-design years that minimum is the mitigation cap, not the SFE. For 2026/27 and 2027/28 the $K = 3$ equilibrium is at the VRR Point (a) cap ($329 and $333), which sits below any plausible TPS mitigation cap for combined-cycle-anchored capacity offers; the settlement cap at $325 therefore binds below both the SFE benchmark and the TPS screen, and the counterfactual used for the $612 million two-year transfer (Section 9) uses the SFE rather than the TPS mitigation cap. If TPS mitigation would have bound near $325 in the absence of the settlement—an empirical question that would require endogenizing the TPS test—the SFE-based transfer estimate overstates the true settlement-induced reduction in capacity payments.

Figure 1 and Table 5 report the equilibrium price-cost margin L as a function of the number of strategic sellers K, holding calibration-year parameters fixed. The results reveal three qualitative zones.

Table 5: Equilibrium Price and Price-Cost Margin by K (Baseline Calibration)

	2023/24 (Old VRR)			2025/26 (Old VRR)			2026/27 (New VRR)		
K	p^*	L	cap	p^*	L	cap	p^*	L	cap
2	412	0.64	✓	452	0.67	✓	329	0.54	✓
3	330	0.55		426	0.65		329	0.54	✓
4	215	0.30		261	0.42		228	0.34	
5	168	0.11		184	0.19		177	0.15	
7	151	0.01		152	0.01		152	0.01	
10	151	0.01		151	0.01		151	0.01	

Notes: Prices in $/MW-day. $L = (p^* - c)/p^*$; $c = \$150$/MW-day. ✓ indicates the market clears at the VRR price cap $\bar{p}$.

Table 6 varies the assumed marginal cost c from $100 to $200 per MW-day. The at-cap

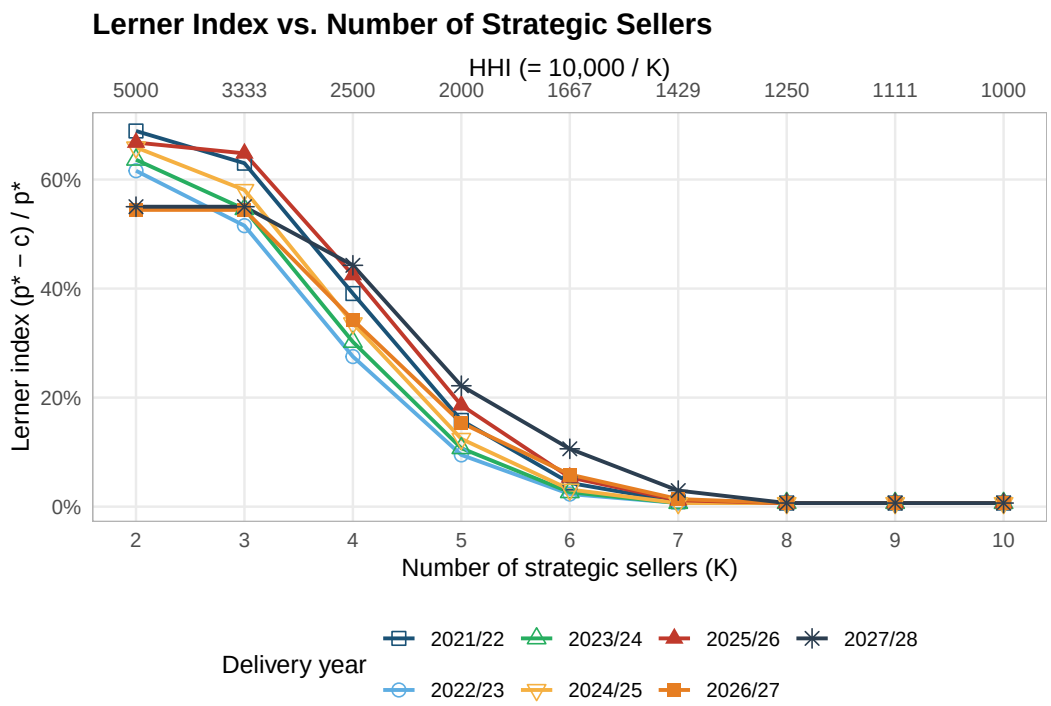

Figure 1: Markups Collapse Between Three and Four Strategic Sellers
Notes: Source: SFE model calibrated to PJM planning parameters for each benchmark year. Shaded region indicates at-cap outcomes where $S_i(\bar{p}) = \bar{q}_i$.

finding is fully robust: p^* exceeds the settlement cap for all cost assumptions shown. The Lerner index in 2026/27 ranges from 0.39 (at $c = \$200$) to 0.70 (at $c = \$100$); interior equilibria show limited price sensitivity to c.

For $K = 2$, the market clears at $\bar{p}$ in all seven delivery years, yielding Lerner indices of 0.54–0.69. The at-cap outcome is also produced by $K = 3$ in 2026/27 (where the VRR cap $\bar{p} = \$329$ is lower, meeting the equilibrium supply function sooner). The transition from $K = 2$ to $K = 3$ has a modest effect on the price-cost margin in 2023/24 (0.64 to 0.55) but almost none in 2025/26 (0.67 to 0.65), because in 2025/26 the supply function remains close to the cap throughout. Adding a fourth seller reduces the price-cost margin by 33–45 percentage points relative to $K = 3$, shifting the 2023/24 equilibrium from \$330 to \$215 and the 2026/27 equilibrium from the cap to \$228. The ODE (11) makes this intuitive: increasing K divides the denominator, forcing $s'(p)$ to be larger in absolute value at any given (p, s) pair, which means firms must offer more capacity at every price below the cap and the equilibrium

Table 6: Equilibrium Price and Price-Cost Margin by Cost Assumption ($K = 3$)

	2023/24 (Old VRR)		2025/26 (Old VRR)		2026/27 (New VRR)	
c (\$/MW-day)	p^*	L	p^*	L	p^*	L
\$100	\$314	0.68	\$422	0.76	\$329	0.70
\$125	\$322	0.61	\$424	0.71	\$329	0.62
\$150	\$330	0.55	\$426	0.65	\$329	0.54
\$175	\$338	0.48	\$428	0.59	\$329	0.47
\$200	\$346	0.42	\$431	0.54	\$329	0.39

Notes: Prices in \$/MW-day. $K = 3$ throughout. For 2026/27, the Holmberg boundary condition is active for all c values shown, so p^* equals the VRR Point (a) cap (\$329/MW-day) in all rows; only the Lerner index varies. Source: SFE model calibrated to PJM planning parameters.

supply function shifts outward. With seven or more strategic sellers, equilibrium prices fall to within 0.5% of marginal cost ($c = \$150$/MW-day) in all seven years, with price-cost margins effectively zero ($L < 0.02$), consistent with the competitive limiting result $p^* \to c$ as $K \to \infty$ implied by equation (11).

The structural implication is direct: the observed 2026/27 at-cap outcome ($p^* = \$329$) is consistent with $K \leq 3$ but inconsistent with $K \geq 4$ (which would predict interior clearing at \$228 or below). This is a model-based inference, not an identification result: under the maintained SFE specification, $K \geq 4$ cannot rationalize the at-cap regime, but K itself is not identified from the auction data.

Figure 2 shows the price-cost margin L as a function of the VRR slope scaling factor ($f = 1$ is the baseline; $f > 1$ is steeper demand).

Steeper VRR demand reduces equilibrium markups by increasing the $D'(p)$ term in the ODE, forcing a more competitive supply function. A doubling of the slope reduces the Lerner index by 20–30 percentage points in interior-clearing years. For 2026/27 at $K = 3$, the outcome remains at the price cap until the slope factor exceeds approximately 1.4, at which point the supply function clears below the cap without an external price constraint.

Figure 3 traces the price-cost margin L as fringe supply $\bar{Q}_f$ varies from 70% to 150% of its calibrated baseline. The fringe captures competitive imports and non-strategic capacity, so this comparative static approximates the effect of expanding CETL or reducing transmission

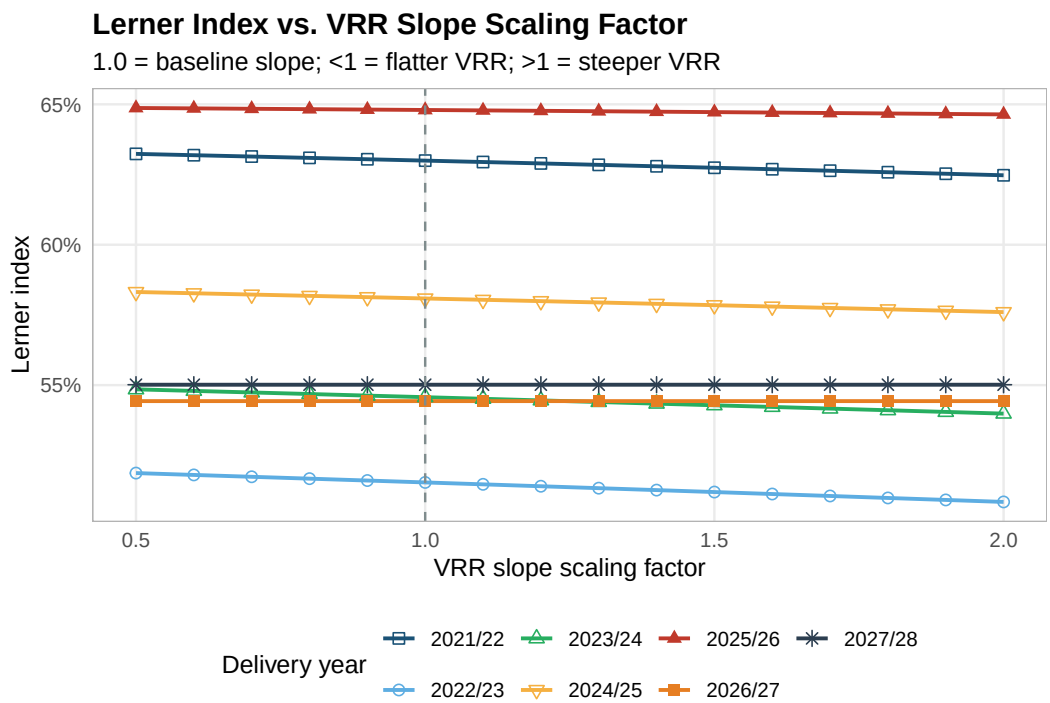

Figure 2: Steeper VRR Demand Reduces Markups and Can Eliminate Cap-Binding
Notes: Source: SFE model with slope factor applied to the VRR quantity increment $(q_b - q_a)$, anchor prices held fixed. Factor = 1 is the calibrated baseline.

congestion.

A 30% increase in fringe supply reduces the price-cost margin by 12–18 percentage points in interior-clearing years. For 2026/27, expanding the fringe at $K = 3$ reduces the margin from 0.54 to roughly 0.30 (a price drop from \$329 to approximately \$215/MW-day), suggesting that expanding import capacity would substantially affect equilibrium pricing even without structural remedies to seller concentration.

Capacity is traded separately in constrained LDAs. For each LDA, I calibrate $\bar{Q}_f = \text{CETL}$ and set each strategic seller's capacity at $\bar{q} = (\text{rel. req.} - \text{CETL})/K$. Table 7 reports the implied Lerner indices for all LDAs with valid CETL data.

Lerner indices range from 0.48 (DEOK in 2025/26) to 0.70 (PS and PS NORTH in 2025/26), with most LDAs above 0.60 in every year. In 2026/27, every LDA clears at or within \$0.50 of the price cap, confirming that the at-cap outcome is not an artifact of RTO aggregation. Figure 5 shows that LDAs with higher import penetration (CETL/rel. req.)

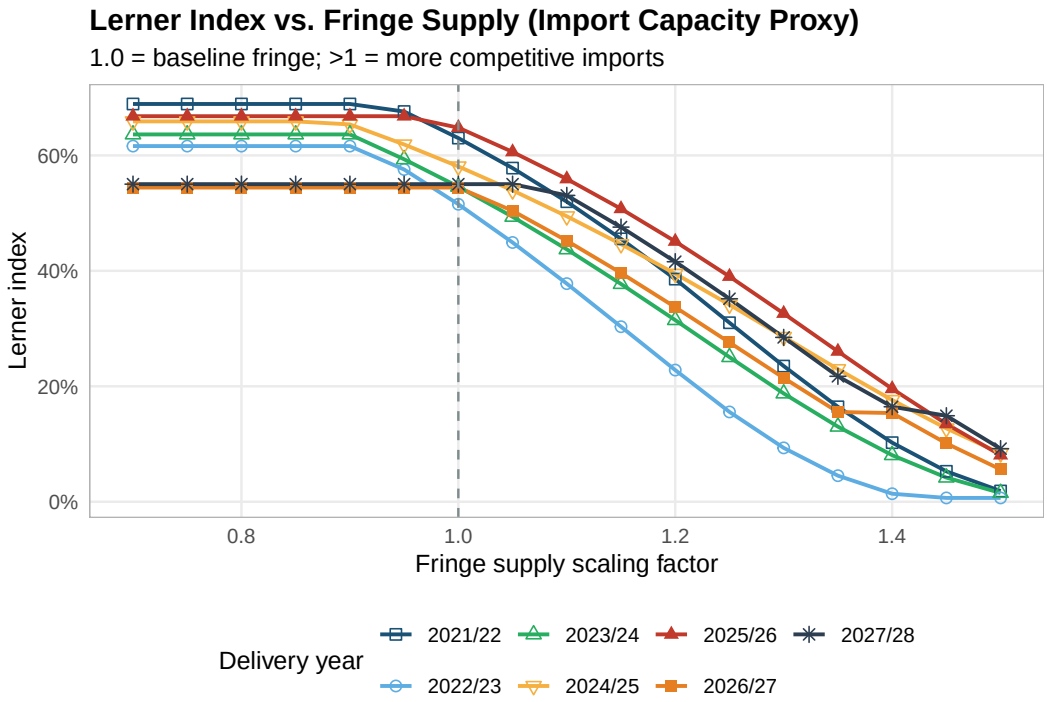

Figure 3: Expanding Competitive Fringe Substantially Reduces Markups
Notes: Source: SFE model with fringe supply $\bar{Q}_f$ scaled from 70% to 150% of the RSI-derived calibrated baseline for each benchmark year.

systematically exhibit lower markups: a 10-percentage-point increase in import penetration is associated with a 4–6-point reduction in the price-cost margin.

7 Cap-Incidence Across Regimes

This section documents which administrative constraint was binding in each of PJM's seven completed Base Residual Auctions: the TPS mitigation cap based on Net ACR, the design-level VRR Point (a) cap, or the Shapiro settlement cap that took effect for the 2026/27 and 2027/28 delivery years. The cap-incidence panel reframes the dramatic price changes observed across the sample period as a regime change in the binding administrative constraint rather than a behavioral shift in seller strategy.

Table 7: SFE Lerner Index by LDA and Delivery Year ($K = 3$, $c = \$150$/MW-day)

LDA	2021/22	2022/23	2023/24	2024/25	2025/26	2026/27	2027/28
ATSI	$0.67^{\dagger}$	$0.57^{\dagger}$	$0.62^{\dagger}$	$0.64^{\dagger}$	0.63	$0.54^{\dagger}$	$0.55^{\dagger}$
ATSI-Cleveland	$0.67^{\dagger}$	0.55	0.61	0.64	0.56	$0.54^{\dagger}$	$0.55^{\dagger}$
COMED	$0.71^{\dagger}$	$0.60^{\dagger}$	$0.65^{\dagger}$	$0.67^{\dagger}$	$0.67^{\dagger}$	$0.54^{\dagger}$	$0.55^{\dagger}$
DEOK	$0.68^{\dagger}$	$0.55^{\dagger}$	$0.60^{\dagger}$	$0.63^{\dagger}$	0.48	$0.54^{\dagger}$	$0.55^{\dagger}$
DOM	–	–	–	–	0.66	$0.54^{\dagger}$	$0.55^{\dagger}$
DPL SOUTH	$0.67^{\dagger}$	$0.58^{\dagger}$	$0.62^{\dagger}$	$0.65^{\dagger}$	0.67	$0.54^{\dagger}$	$0.55^{\dagger}$
EMAAC	$0.68^{\dagger}$	$0.61^{\dagger}$	$0.66^{\dagger}$	$0.68^{\dagger}$	$0.68^{\dagger}$	$0.54^{\dagger}$	$0.55^{\dagger}$
JCPL	–	–	–	–	–	$0.54^{\dagger}$	$0.55^{\dagger}$
MAAC	$0.66^{\dagger}$	$0.59^{\dagger}$	$0.64^{\dagger}$	$0.66^{\dagger}$	$0.67^{\dagger}$	$0.54^{\dagger}$	$0.55^{\dagger}$
PEPCO	$0.65^{\dagger}$	$0.61^{\dagger}$	$0.63^{\dagger}$	$0.65^{\dagger}$	–	$0.54^{\dagger}$	–
PL	$0.67^{\dagger}$	$0.60^{\dagger}$	$0.64^{\dagger}$	$0.66^{\dagger}$	0.65	$0.54^{\dagger}$	$0.55^{\dagger}$
PS	$0.70^{\dagger}$	$0.63^{\dagger}$	$0.67^{\dagger}$	$0.69^{\dagger}$	$0.70^{\dagger}$	$0.54^{\dagger}$	$0.55^{\dagger}$
PS NORTH	$0.70^{\dagger}$	$0.63^{\dagger}$	$0.67^{\dagger}$	$0.69^{\dagger}$	$0.70^{\dagger}$	$0.54^{\dagger}$	$0.55^{\dagger}$
SWMAAC	$0.63^{\dagger}$	$0.59^{\dagger}$	$0.59^{\dagger}$	$0.62^{\dagger}$	$0.68^{\dagger}$	$0.54^{\dagger}$	$0.55^{\dagger}$
BGE	$0.63^{\dagger}$	$0.56^{\dagger}$	$0.55^{\dagger}$	$0.58^{\dagger}$	0.68	0.54	$0.55^{\dagger}$
DAYTON	$0.68^{\dagger}$	$0.56^{\dagger}$	–	$0.62^{\dagger}$	–	–	–

Notes: $K = 3$ strategic sellers per LDA. $c = \$150$/MW-day (CC technology-average ACR, 2025 IMM SotM). Market structure calibrated using $\bar{Q}_f$ = CETL (competitive imports) and $\bar{q}$ = (rel. req. − CETL)/K (strategic sellers cover local gap). † Market clears at or within \$0.50 of the VRR price cap.

Table 8: Cap-Incidence Panel: Which Administrative Constraint Was Binding, PJM Base Residual Auctions 2021/22–2027/28 (RTO Level)

Year	VRR Design	Clearing price	Net ACR	VRR Pt (a)	Settle. cap	Reserve excess IRM (MW)	TPS mitig.	Binding cap
2021/22	Old	\$140	\$150	\$482	—	—	—	None
2022/23	Old	\$50	\$150	\$391	—	—	0.0%	None
2023/24	Old	\$34	\$150	\$412	—	+5,687	7.3%	TPS active
2024/25	Old	\$29	\$150	\$440	—	+3,278	2.2%	TPS active
2025/26	Old	\$270	\$150	\$452	—	-205	5.5%	TPS active
2026/27	New	\$329	\$150	\$329	\$325	-492	6.3%	Settlement
2027/28	New	\$333	\$150	\$333	\$325	-6,516	1.0%	Settlement

Notes: Prices in nominal \$/MW-day UCAP. Net ACR is the IMM-published sector-average Avoidable Cost Rate (\$149.32, rounded to \$150) used as the default mitigation offer cap for resources subject to the Three Pivotal Supplier (TPS) test. VRR Pt (a) is the design-level price cap on the unmodified Variable Resource Requirement curve. Settlement cap is the Shapiro/PJM \$325/MW-day UCAP cap that took effect for the 2026/27 and 2027/28 delivery years. Reserve excess IRM (MW) is committed UCAP minus IRM-implied required reserves (negative = shortfall). TPS mitigation percentage is the share of generation resources that submitted Capacity Performance offers and had unit-specific Market

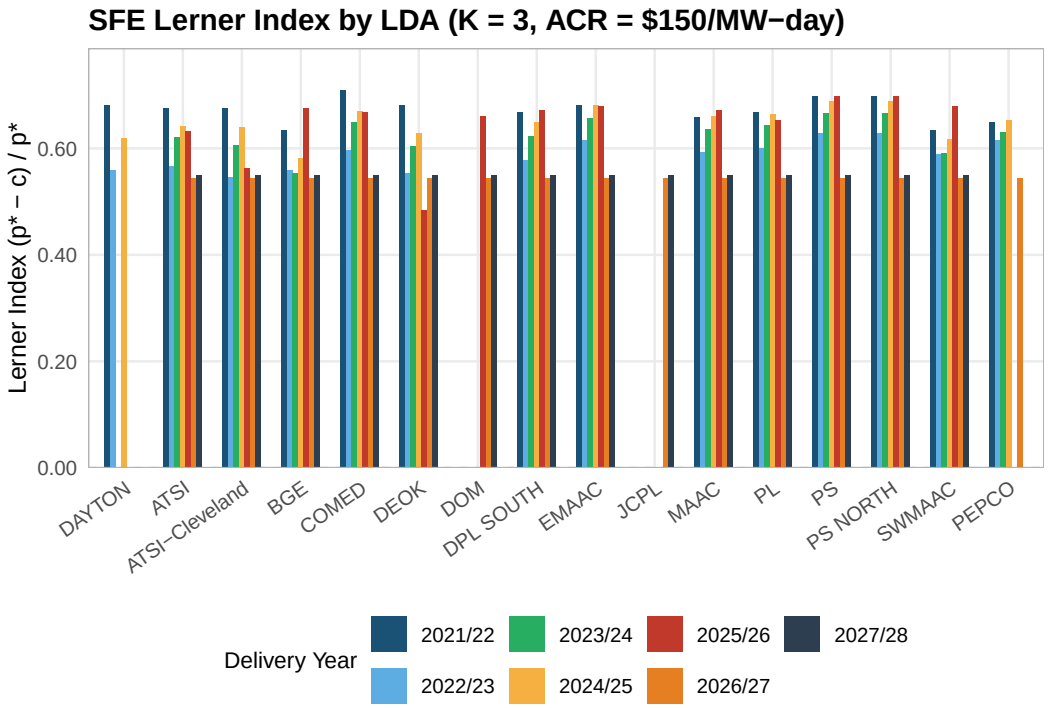

Figure 4: All Locational Markets Exhibit High Markups, Especially Import-Constrained LDAs
Notes: Source: SFE model calibrated to PJM planning parameters and CETL data by LDA. †: market cleared at or within $0.50 of the VRR price cap. LDAs ordered by 2026/27 margin.

Three regimes are visible in Table 8, distinguished by the joint pattern of clearing price, capacity margin, and IMM-applied mitigation activity.

The first regime, covering 2021/22 through 2024/25, is one of comfortable supply margins and cost-anchored clearing. The IMM-published reserve margin in excess of the Installed Reserve Margin (IRM) was strongly positive in 2023/24 and 2024/25 (+5,687 MW and +3,278 MW respectively; data for 2021/22 and 2022/23 are not in the IMM panel I assembled), and clearing prices ranged from $29 to $140/MW-day, well below both the sector-average Net ACR and the design-level VRR Point (a) cap of $391–$482/MW-day. TPS mitigation was applied to a small share of generation resources (0.0% in 2022/23, 7.3% in 2023/24, 2.2% in 2024/25), but with the marginal cleared offer well below any active cap, mitigation operative for a minority of supra-marginal resources did not pin the clearing outcome. The classification "TPS active" for 2023/24 and 2024/25 records the IMM mitigation activity without claiming that the mitigation cap determined the clearing price; "None" for 2021/22

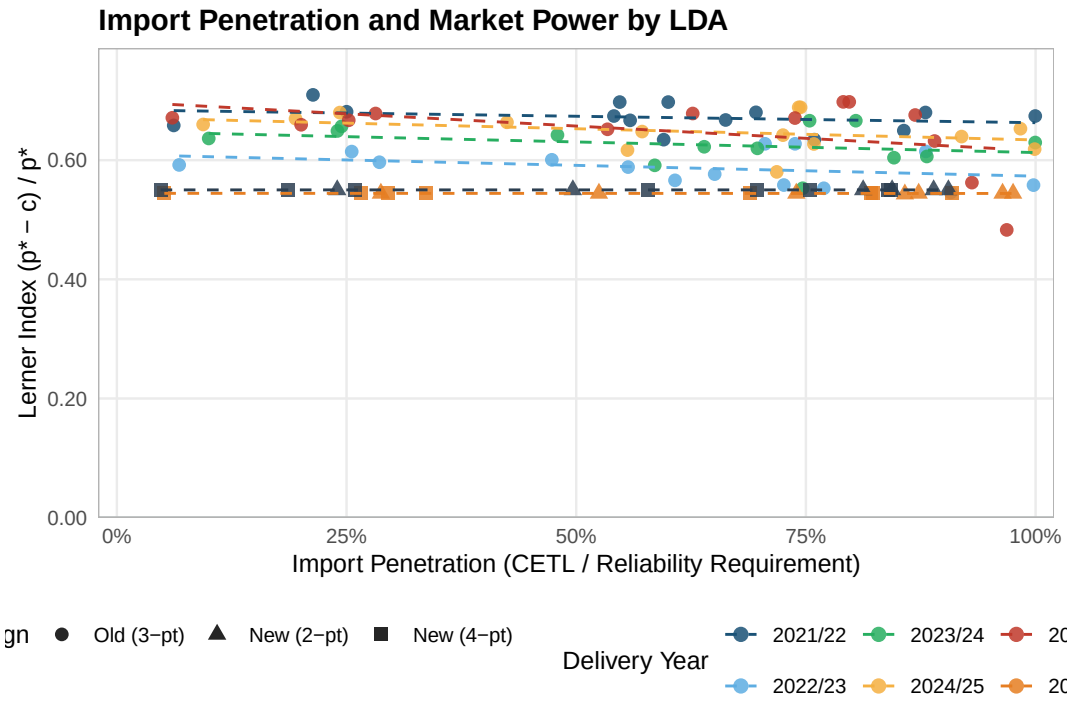

Figure 5: Higher Import Penetration Systematically Reduces Local Markups
Notes: Source: SFE model calibrated to PJM planning parameters and CETL data by LDA. Each point is one LDA-year observation; dashed lines are ordinary least squares (OLS) trends by year.

and 2022/23 records a slack market in which mitigation was either inactive or not reported in locally available IMM data.

The second regime, in 2025/26, is a transition. The reserve margin in excess of IRM turned slightly negative for the first time (−205 MW), the clearing price jumped from $29 in the preceding auction to $270/MW-day, and TPS mitigation rose to 5.5% of resources. Yet $270 remains far below the design Point (a) of $452/MW-day, so no administrative cap was binding the clearing. 2025/26 is the auction in which the underlying supply–demand balance tightened materially without any cap yet binding the headline outcome.

The third regime, comprising 2026/27 and 2027/28, is one of cap-binding clearing under negative reserve excess. Both auctions cleared at the new-design VRR Point (a) value, which under the Shapiro/PJM settlement embodies the agreed maximum price (the nominal $325/MW-day target translates to $329.17 and $333.44 in 2026/27 and 2027/28 respectively after year-specific accreditation and Net CONE adjustments). The reserve margin in excess

of IRM was −493 MW in 2026/27 and −6,517 MW in 2027/28, with the IMM declaring both auctions "not competitive" (Monitoring Analytics, LLC, 2026).[2] The drop in TPS mitigation application from 6.3% in 2026/27 to 1.0% in 2027/28 is consistent with the settlement cap moving sufficiently close to the underlying offer distribution that fewer individual offers exceeded their unit-specific Net ACR caps.

The cap-incidence panel offers a direct answer to a recurring question about the post-settlement auctions: did seller behavior change between the slack and tight regimes, or did the binding administrative constraint change while the underlying competitive process remained similar? The data favor the latter interpretation. Across the sample, the SFE machinery developed in Section 4 predicts strategic markups that are far above observed slack-year clearings and mechanically equal to the cap in tight years. The honest reading is that the SFE identifies the at-cap versus interior *regime*, conditional on supply margin and concentration, but does not independently identify the dollar clearing price in either regime. What identifies clearing in the data is the lowest binding administrative constraint: cost in slack years, the settlement cap in tight years. The apparent dramatic price change between regimes is therefore at least in part a regime change in which constraint binds, not a structural shift in strategic conduct.

This reading does not contradict the IMM's substantive assessment. The IMM attributes the underlying scarcity in 2026/27 and 2027/28 to data center load growth rather than to any change in seller behavior or to the settlement itself (Monitoring Analytics, LLC, 2026). The contribution of the cap-incidence panel is to make that interpretation visible at a glance and to anchor the more detailed counterfactual discussion in Section 9 in observable cap-incidence facts rather than in model-implied prices that are mechanically tied to whichever cap is operative.

[2]The IMM SoM 2025 reports the 2026/27 reserve shortfall in two forms: the panel-consistent "Reserve cleared in excess of IRM UCAP" figure of −493 MW used here, and a parallel calculation reported as "208.7 MW short of the required reserve level of 21,561.9 MW" computed against a different reserve baseline. Both characterize a marginal shortfall.

8 Allocative Implications of the Price Band

The settlement cap's allocative significance depends on its proximity to the investment threshold. In 2026/27, Net CONE is \$212.14/MW-day; the settlement cap at \$325/MW-day exceeds it by 53%, so the cap alone does not prevent new entry from being financially viable. The more substantive concern is informational: because the cap binds in consecutive years (2026/27 and 2027/28), potential entrants observe the cap rather than the unconstrained equilibrium price and cannot distinguish between a market that would have cleared at \$333 and one that would have cleared far above \$325. This uncertainty reduces the informativeness of the capacity price for investment decisions (Hogan, 2005; Joskow, 2008), a concern amplified in constrained LDAs where local Net CONE values exceed the settlement cap (e.g., PS and PS NORTH Net CONE = \$369/MW-day in 2026/27).

The \$175/MW-day floor operates differently: PJM describes it as protecting suppliers from price collapse in over-supplied conditions (Bresler and Keech, 2025), which I read as an investment-adequacy mechanism rather than a cost-reduction measure. At the RTO level, the floor falls below Net CONE (\$212.14) and does not guarantee full cost recovery. Combined with the cap, the settlement creates a price band (\$175–\$325) that is neither the competitive equilibrium ($c \approx$ \$150) nor the SFE benchmark ($p^* \approx$ \$329–\$333). For $K = 3$, the SFE equilibrium falls above the band, so the cap truncates the price signal rather than confirming it. A formal welfare accounting— comparing deadweight losses from the truncated investment signal against consumer savings from lower cleared prices—is outside the scope of this paper; the observation here is that the cap is close enough to Net CONE to leave entry incentives broadly intact, and the remaining allocative concern is informational rather than direct cost recovery.

9 Evaluating the $21 Billion Claim

Pennsylvania's January 28, 2025 announcement stated that the settlement would save consumers "over $21 billion over the next two years" (Office of Governor Josh Shapiro, 2025). The arithmetic is straightforward: (counterfactual price − $325) × cleared quantity × 2 years. Shapiro's stated counterfactual is "over $500/MW-Day" (Office of Governor Josh Shapiro, 2025), consistent with the pre-settlement VRR Point (a) price visualized in PJM's settlement consultation (Bresler and Keech, 2025). But the VRR Point (a) price is not a strategic equilibrium price. It is the maximum of the administrative demand schedule: the price at which the VRR curve becomes perfectly inelastic. In equilibrium, the capacity market clears where supply meets the VRR schedule, which, as Section 6 shows, may occur anywhere in the range $[c, \bar{p}]$ depending on market structure. The VRR counterfactual therefore uses an administrative ceiling rather than a market-clearing benchmark.

The SFE model provides a structural alternative: p^*, the equilibrium price under strategic bidding against the *unmodified* VRR curve. In 2026/27, $p^* = \$329.17$/MW-day; in 2027/28, $p^* = \$333.44$/MW-day. Both equilibria fall at the respective VRR Point (a) caps because the Holmberg boundary condition is active for $K \leq 3$, but they are substantially below the $500 figure in Shapiro's claim.

Table 9 compares the implied revenue transfer under two counterfactual scenarios for the 2026/27 and 2027/28 delivery years, using RTO-level cleared quantities from the published BRA results.

Table 9: Revenue Transfer Under Two Counterfactual Scenarios (RTO Level)

	Counterfactual price		Annual revenue transfer		Two-year
Scenario	2026/27	2027/28	2026/27	2027/28	total
VRR Point (a) [Shapiro]	$500	$500	$8.57B	$8.59B	$17.16B
SFE equilibrium [this paper]	$329	$333	$0.20B	$0.41B	$0.61B
Settlement cap	$325	$325	—	—	—

Notes: All prices in $/MW-day UCAP. Annual revenue transfer = $(p_{\text{counterfactual}} - \$325) \times Q \times 365$, where $Q = 134{,}205$ MW (2026/27) and $Q = 134{,}478$ MW (2027/28). Source: PJM BRA clearing data and SFE model.

Using \$500/MW-day as the counterfactual and RTO-level quantities, the two-year revenue transfer is \$17.16 billion. The stated \$21 billion figure includes contributions from constrained LDAs. The structural SFE estimate implies a two-year revenue transfer of approximately \$612 million at the RTO level—substantially smaller—because the 2026/27 strategic equilibrium falls near the design ceiling, not because the settlement eliminates large rents. The two counterfactuals differ not because one is wrong, but because they answer different questions: the VRR counterfactual measures the gap between the design-level cap and the settlement cap, while the SFE counterfactual measures the reduction in the predicted clearing price attributable to the price band. Both estimates condition on which Point (a) represents the pre-settlement benchmark; as Section 9 notes, a counterfactual using the pre-settlement VRR Point (a) rather than the new-design value would yield a larger SFE-based transfer under the scarcity conditions observed in 2026/27 and 2027/28.

The findings rest on maintained assumptions whose validity cannot be fully tested. First, I assume $K = 3$ symmetric sellers with identical costs. Actual PJM suppliers differ in technology, cost structure, and contract positions, so the symmetric Lerner index should be interpreted as a central estimate, not a precise measurement. Second, the Holmberg boundary condition is derived for stochastic-demand markets; in the deterministic BRA, it is imposed as an equilibrium selection criterion, and alternative selection rules would yield different equilibrium prices. Third, the static model treats capacity as exogenous and cannot quantify entry distortions from the settlement. A complete savings calculation should account for retirements accelerated by lower prices and entry decisions informed by the observed 2026/27 and 2027/28 clearing prices; the \$21 billion figure and the SFE estimate both hold quantity fixed. Fourth, even if greenfield entry is infeasible within the 10-month lead time of the 2026/27 BRA, auction prices inform decisions beyond direct construction. The clearest such margin is retirement deterrence (Cramton and Stoft, 2005); additional margins— demand response registration, bilateral contracting, and forward expectations for 2028/29 and beyond—are plausibly affected as well, so the informational cost of suppressing

the price signal extends beyond the settlement's two-year window. Fifth, I do not model TPS mitigation endogenously; the gap between predicted and actual prices in old-design years suggests mitigation effects the model does not capture. If TPS mitigation would have produced offer caps near the settlement cap in 2026/27 absent the settlement, the SFE-based transfer of \$612 million overstates the settlement's true reduction in payments; the observed old-design gap between p^* and p_{actual} cannot be used to rule this out, because TPS screening in those years occurred with capacity margins of 9.9% and 11.7% where the test was much more likely to be triggered than in the at-cap 2026/27 and 2027/28 auctions, where RSI_3 was binding but TPS application is regime-dependent.

Sixth, both new-design auctions cleared below their reliability requirements: the 2026/27 RTO auction cleared 314 MW short (-0.23%) and the 2027/28 auction cleared 6,623 MW short (-4.69%), as shown in Table 4. This overdetermines the at-cap outcome. The Klemperer–Meyer intuition that makes Holmberg selection economically informative—rivals' capacity constraints bind at $\bar{p}$ because sellers strategically withhold, making residual demand perfectly inelastic—is observationally indistinguishable from physical scarcity, under which aggregate offered capacity falls short of the quantity required at any price below the cap and the market clears at $\bar{p}$ regardless of K. The $K = 4$ counterfactual of interior clearing at \$228/MW-day for 2026/27 (Table 5) presupposes that four symmetric strategic sellers plus the fringe can jointly deliver $D(\$228)$; the observed shortfall is direct evidence that this supply-sufficiency condition fails, in which case $K \geq 4$ would also predict at-cap clearing. The "consistent with $K \leq 3$ but inconsistent with $K \geq 4$" inference in Section 6 therefore conditions on a supply-sufficiency assumption that the 2026/27 and 2027/28 data contradict. By the same mechanism, the \$612 million transfer estimate is sensitive to the choice of pre-settlement Point (a): if the relevant counterfactual VRR has a Point (a) closer to \$500 than to the new-design \$329 (consistent with the pre-settlement design visualized in PJM's settlement slides and with Shapiro's stated counterfactual), the SFE under scarcity would also clear near \$500, substantially narrowing the apparent gap with the \$17.16 billion figure

in Table 9. The qualitative finding that the settlement cap binds under concentrated supply is preserved; the quantitative claims—both "$K \leq 3$ is identified" and the specific \$612 million versus \$17.16 billion comparison—are more contingent on maintained assumptions than the baseline table alone suggests.

These limitations bound the precision of the quantitative estimates but do not affect the qualitative finding that the settlement cap is binding under concentrated supply.

10 Conclusion

Three findings emerge from the calibrated SFE analysis. First, the settlement cap is binding under concentrated supply: with $K \leq 3$ pivotal suppliers, the SFE equilibrium for 2026/27 falls at the VRR Point (a) cap of \$329.17/MW-day, exceeding the settlement cap $\bar{p}_S = \$325$ by \$4.17/MW-day at the RTO level and by \$115–\$141/MW-day in constrained LDAs. Second, the \$21 billion savings figure depends on the counterfactual: using the SFE equilibrium rather than the VRR Point (a) price yields a two-year RTO transfer of approximately \$612 million, substantially smaller than the stated figure; both magnitudes are sensitive to which Point (a) the pre-settlement VRR is taken to fix, and the at-cap outcome in 2026/27 and 2027/28—each of which cleared below its reliability requirement—is consistent with both the SFE mechanism and physical scarcity (Section 9). Third, structural channels can reduce prices without truncating the signal: CETL expansion, VRR slope steepening, and entry sufficient to reach $K \geq 4$ each lower the SFE equilibrium through concentration or fringe supply rather than direct price truncation.

The model's contribution is evaluative rather than prescriptive. A temporary cap can reduce payments, but the comparative statics show that the underlying concentration and transmission conditions generating high equilibrium prices are what place the equilibrium at the cap in the first place. The structural channels identified in the comparative statics (CETL expansion, VRR slope, entry) would reduce equilibrium markups through market

structure rather than by truncating the price signal; the model does not rank reforms or quantify their welfare consequences, and whether any particular change is warranted depends on regulatory objectives and cost–benefit considerations outside its scope. Natural extensions include an asymmetric-firm SFE, a dynamic model connecting markups to capacity adequacy, and an empirical structural model recovering firm-level costs from offer data.

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